Q1. Dependencies used and why?

- Sqflite – Used to interact with sqlite database
  – Provide Methods for executing SQL queries (CRUD)
- Path – Helps to construct file system path in Platform independent way.
- Path_provider – Used to locate device storage directories (Eg. where database file is stored.
- intl (optional) used for formating dates & timestamps

Why Sqlite Necessary to store data locally?

> It allows offline data storage (No internet)
> It is lightweight and embedded.
> It supports structured data storage using tables
> It enables persistent storage.

Q2. Handling database exceptions

a) Handling Database Not initialized
Problem: Database May be Null or Not yet created

Solution code

```
Future < database > get database asyncu {
  if (_database != Null) return _database;
  _database = await _initDB ('posts.db');
  return _database;
```

b) Handling CRUD Errors
problem: SQL operation May Fail (Invalid query)
Solution (use Try and catch)

c.) Handling Invalid or corrupted data
problem: – Empty field, wrong datatype
– Corrupted records.
Solution to
– Input Validation
– Safe Data Mapping

Q3. Explain Sqlite in Flutter

a) Database VS Table

- Database is container that holds data of single .db files stored on device Ex: post_Manager.db
- Table structured grid of rows and columns inside the database.
- Relationship one database can contain Many Tables.

b). CRUD operations

C (Create)	Insert New data	like	add Post
R (Read)	Retrieve data	like	View all Post
U (update)	Modify data	like	Edit post
D (Delete)	Remove data	like	Delete post

Code:

```
db.insert('posts', post.toMap());  // create
await db.querry('posts')           // Read
await db.update('posts', post.toMap());  // update
await db.delete('posts', where 'id = ?');  // delete
```

c) How flutter interacts with Database asynchronously

> flutter uses asynchronous Programming to avoid Freezing the UI.

Key Concepts
- Future
- Async / await

Example:

Code

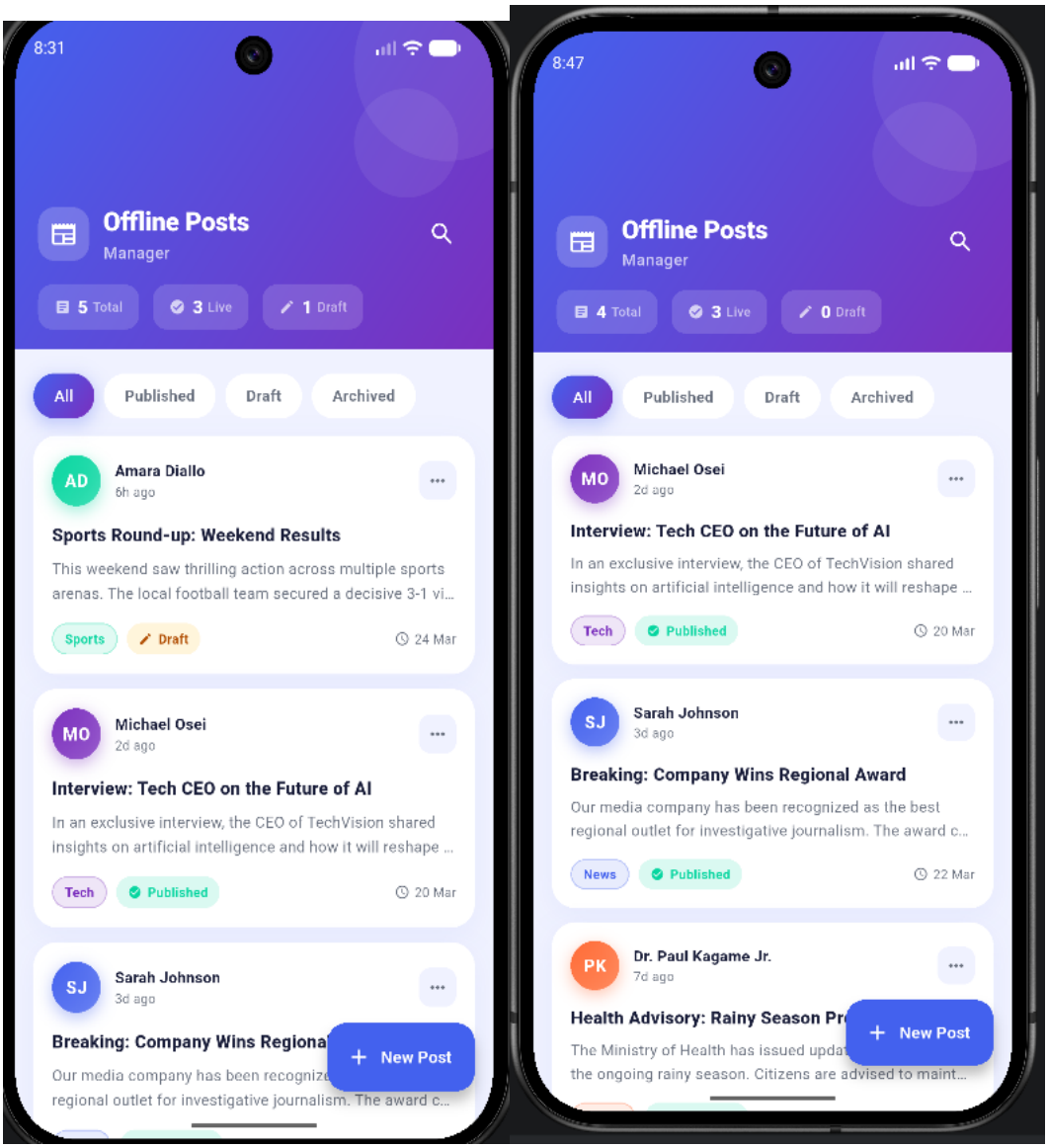
```
Future<List<post>> getallpost() async {
  Final db = await database;
  Final result = await db.querry('post');
  return result.Map(e) => Post.FromMap(e).toList();
}
```

Why async?
- Database operation take time
- Prevent UI lag or Freezing
- Ensures smooth user Experience
-

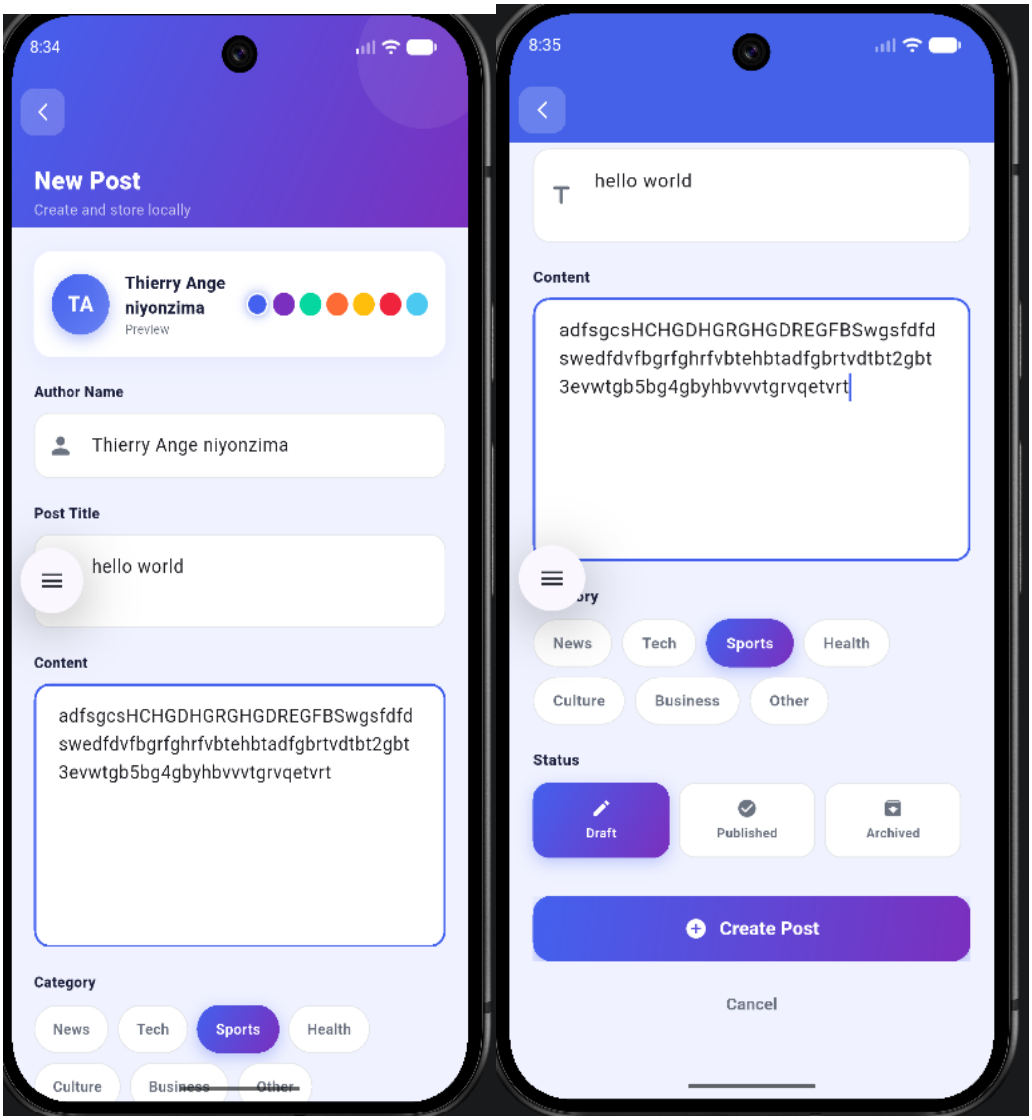
! Without async app became unresponsive

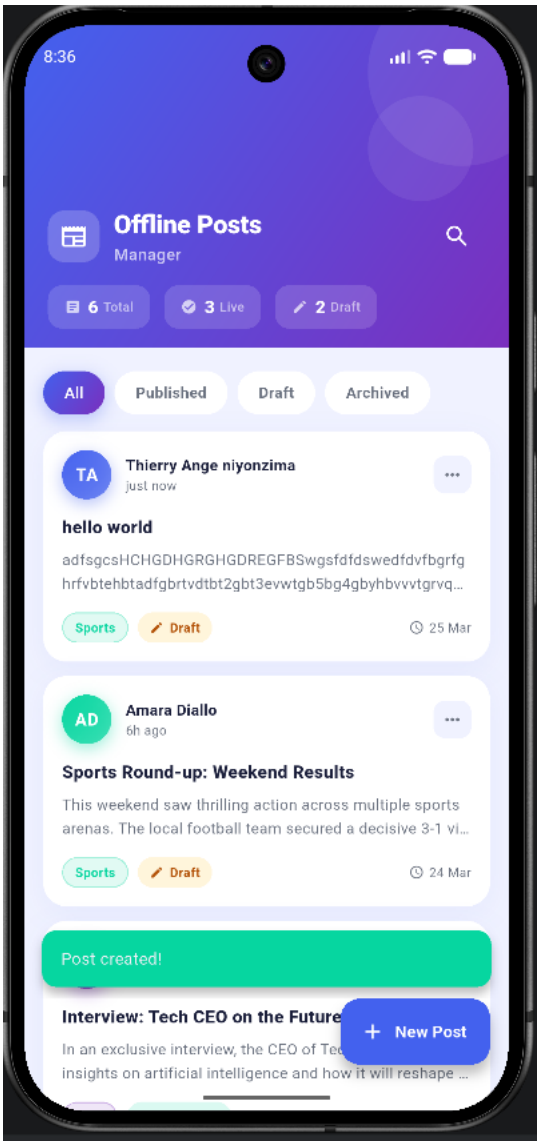
Application screenshot

View all post

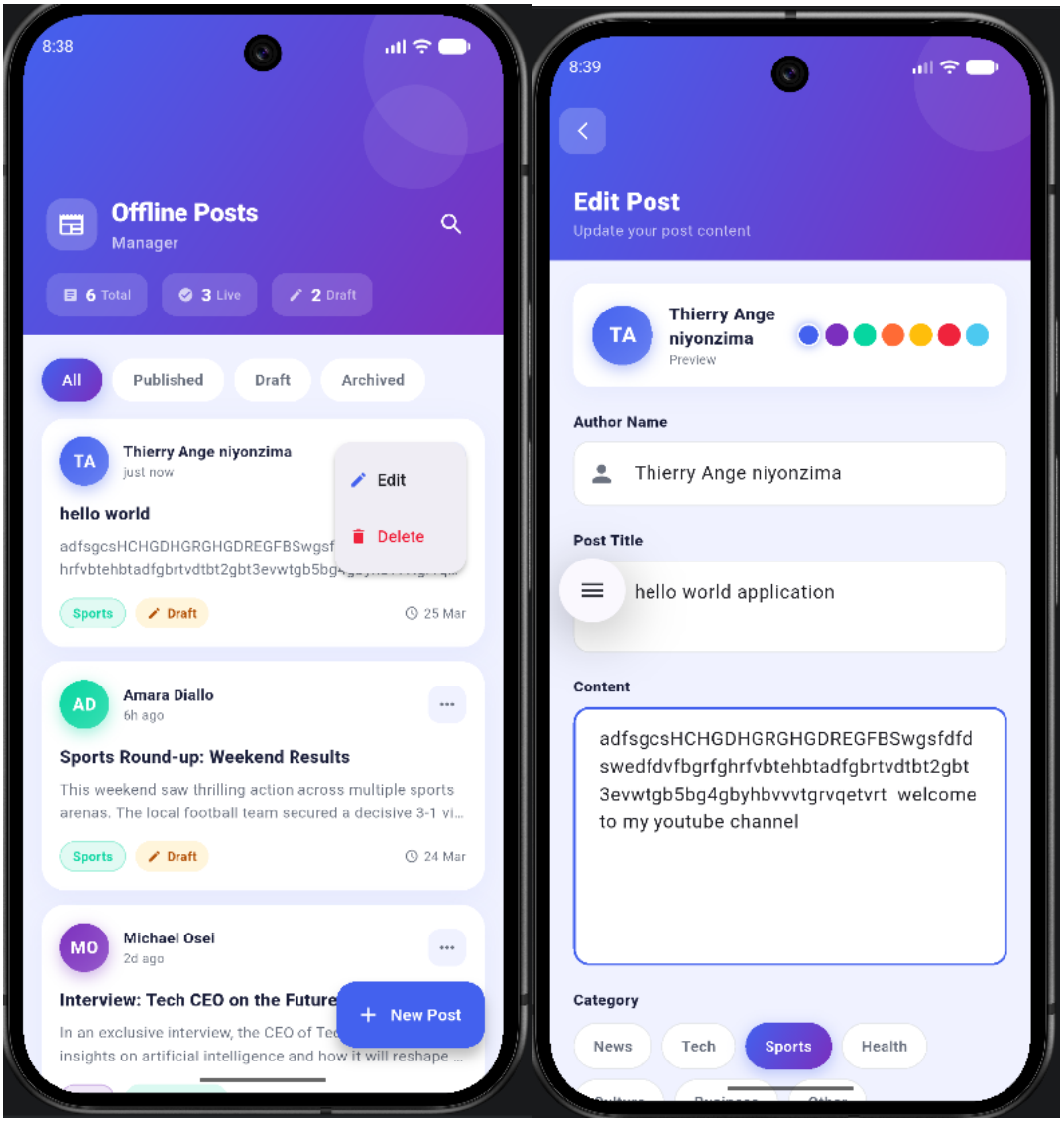


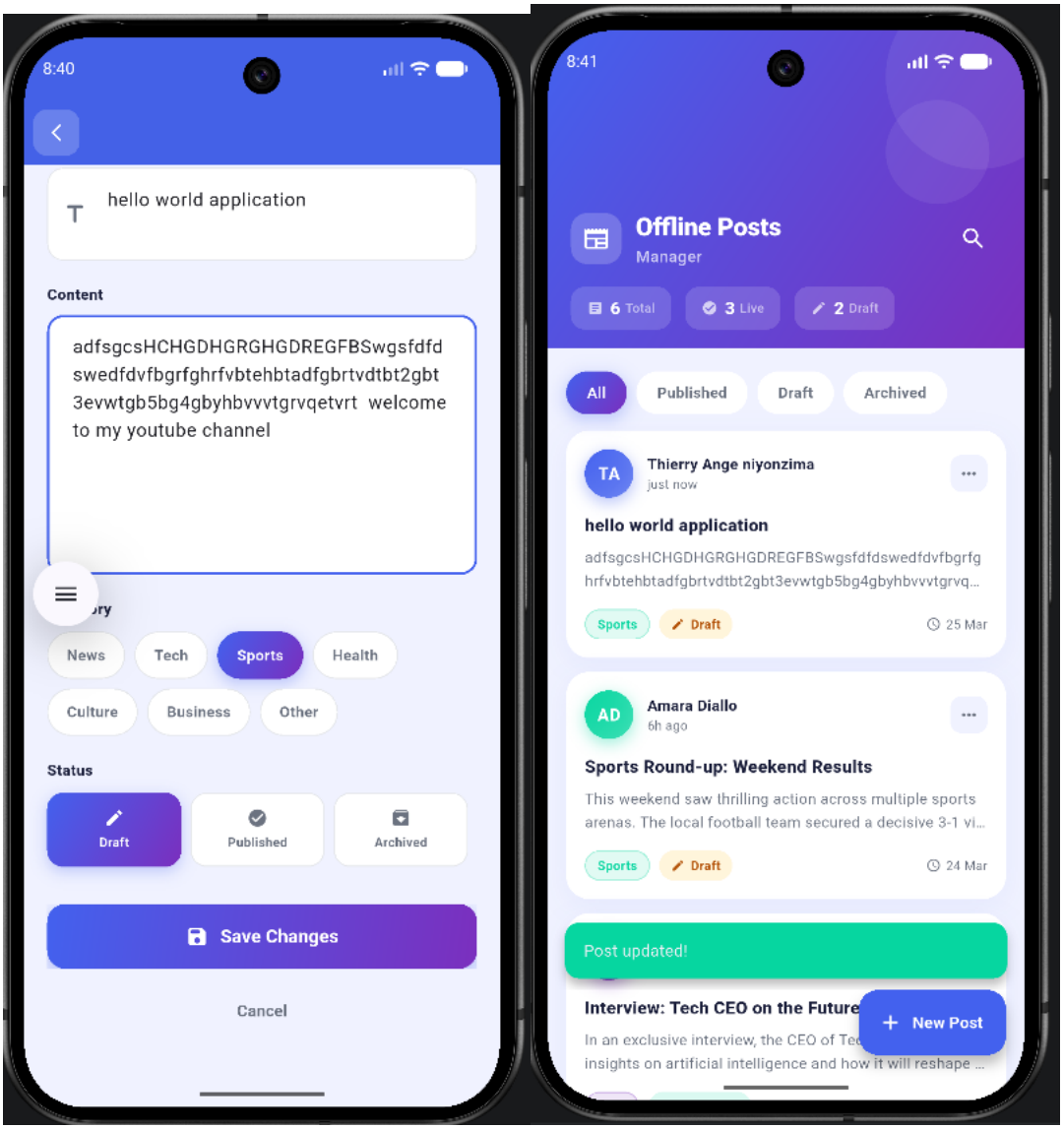
Creating new post


8:34
New Post
Create and store locally
TA
Thierry Ange niyonzima
Preview
Author Name
Thierry Ange niyonzima
Post Title
hello world
Content
adfsgcsHCHGDHGRGHGDREGFBSwgsfdfd swedfdvfbgrfghrfvbtehbtadfgbrtvdtbt2gbt 3evwtgb5bg4gbyhbvvvtgrvqetvrt
Category
News
Tech
Sports
Health
Culture
Business
Other
8:35
hello world
Content
adfsgcsHCHGDHGRGHGDREGFBSwgsfdfd swedfdvfbgrfghrfvbtehbtadfgbrtvdtbt2gbt 3evwtgb5bg4gbyhbvvvtgrvqetvrt
News
Tech
Sports
Health
Culture
Business
Other
Status
Draft
Published
Archived
Create Post
Cancel



Editing post


8:38
Offline Posts
Manager
6 Total
3 Live
2 Draft
All
Published
Draft
Archived
TA
Thierry Ange niyonzima
just now
Edit
Delete
hello world
adfsgcsHCHGDHGRGHGDREGFBSwgsf
hrfvbtehbtadfgbrtvdtbt2gbt3evwtgb5bg
Sports
Draft
25 Mar
AD
Amara Diallo
6h ago
Sports Round-up: Weekend Results
This weekend saw thrilling action across multiple sports arenas. The local football team secured a decisive 3-1 vi...
Sports
Draft
24 Mar
MO
Michael Osei
2d ago
Interview: Tech CEO on the Future
In an exclusive interview, the CEO of Te
insights on artificial intelligence and how it will reshape ...
New Post
8:39
Edit Post
Update your post content
TA
Thierry Ange niyonzima
Preview
Author Name
Thierry Ange niyonzima
Post Title
hello world application
Content
adfsgcsHCHGDHGRGHGDREGFBSwgsfdfd swedfdvfbgrfghrfvbtehbtadfgbrtvdtbt2gbt 3evwtgb5bg4gbyhbvvvtgrvqetvrt welcome to my youtube channel
Category
News
Tech
Sports
Health


8:40
hello world application
Content
adfsgcsHCHGDHGRGHGDREGFBSwgsfdfd swedfdvfbgrfghrfvbtehbtadfgbrtvdtbt2gbt 3evwtgb5bg4gbyhbvvvtgrvqetvrt welcome to my youtube channel
News
Tech
Sports
Health
Culture
Business
Other
Status
Draft
Published
Archived
Save Changes
Cancel
8:41
Offline Posts
Manager
6 Total
3 Live
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Draft
24 Mar
Post updated!
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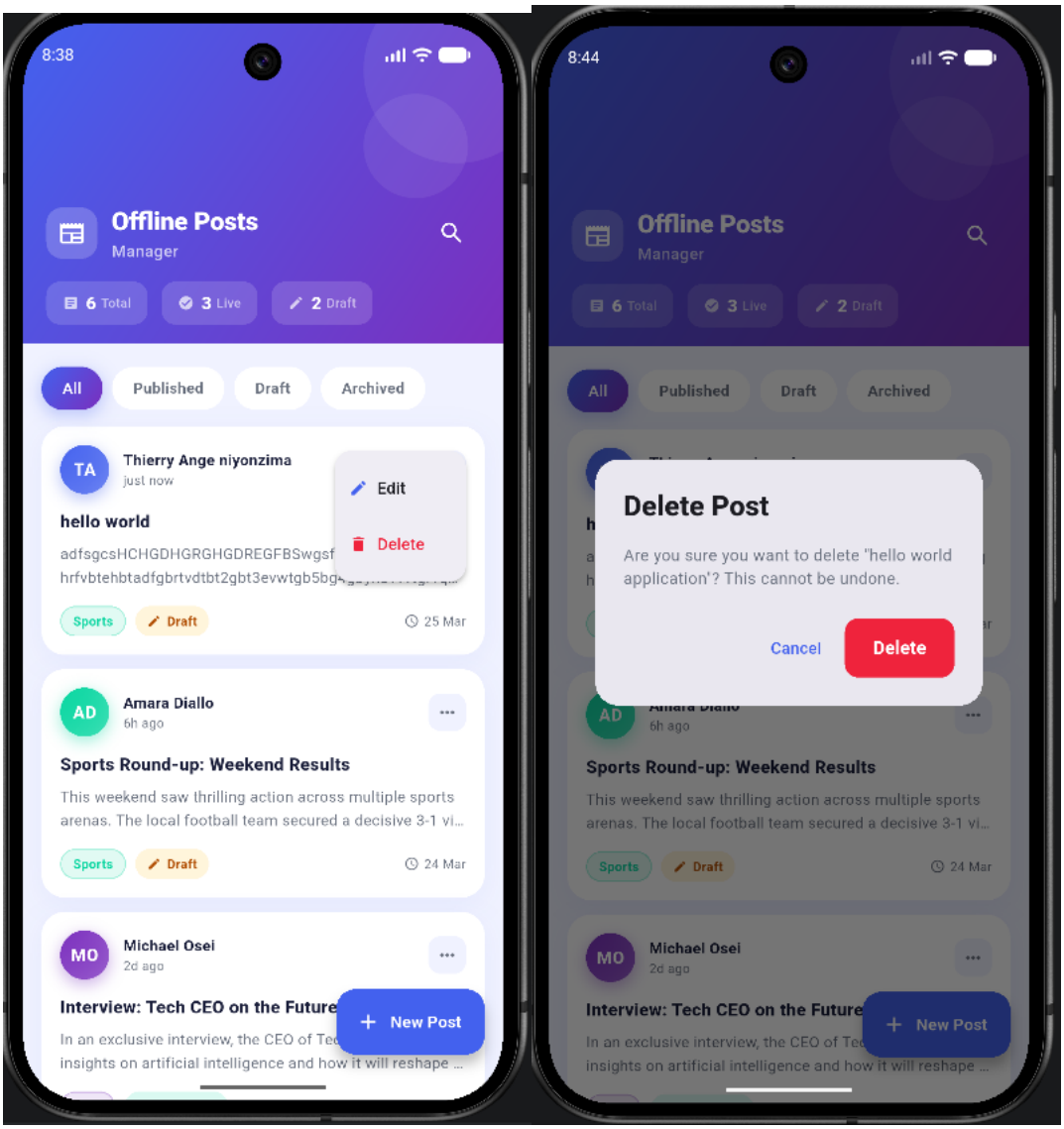
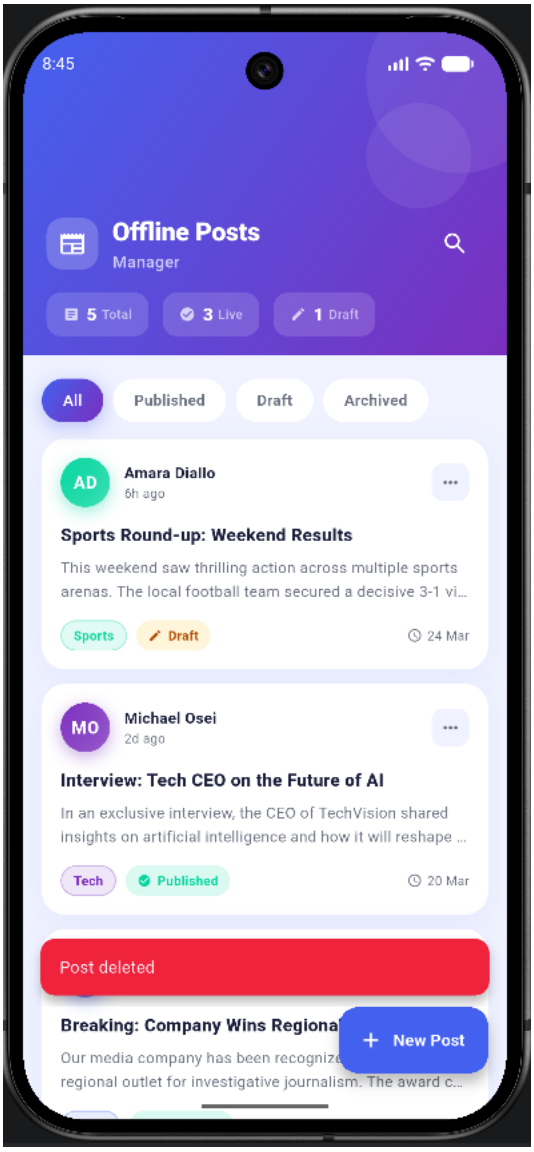
insights on artificial intelligence and how it will reshape ...
New Post

Deleting post




8:45
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Manager
5 Total
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Tech
Published
20 Mar
Post deleted
Breaking: Company Wins Regiona
Our media company has been recogniz
regional outlet for investigative journalism. The award c...
New Post